Datathon Problem Statement: The Financial Resilience Stress Test

1. Context & Background

Financial health isn't just about how much money people have; it is about how active that money is. The **Credit-Deposit Ratio (CDR)** is a critical banking metric that indicates how much of the money deposited in a district is being reinvested there as loans to fuel growth.

- **High CDR:** The district is economically vibrant; capital is being put to work (loans are flowing).
- **Low CDR:** The district is suffering from "Capital Flight"—money is being saved but not invested back into the local community.

Simultaneously, the **National Family Health Survey (NFHS)** provides a granular view of the "Social Health" of a district—measuring literacy, sanitation, internet usage, and lifestyle.

The Research Question: *Can we predict the financial vibrancy (CDR) of a district solely based on its social and health indicators?*

2. The Challenge

Your task is to build a Machine Learning model that predicts the **Credit-Deposit Ratio (CDR)** of an Indian district using **only** its socio-economic data (NFHS).

This is not just a modeling challenge; it is a **Data Engineering** challenge. You are provided with two distinct, real-world datasets from different government bodies. They do not talk to each other. Your first hurdle is to make them communicate.

3. Dataset Description

You have been provided with two raw datasets:

- **Dataset A (Financial):** Sourced from the **RBI**. Contains data on "Aggregate Deposits" and "Bank Credit" for districts across India.
    - *Note:* Real-world data often contains metadata rows and formatting inconsistencies.
- **Dataset B (Social):** Sourced from the **NFHS**. Contains 100+ indicators on population, health, literacy, and infrastructure for each district.

4. Key Objectives

Phase 1: Data Engineering (The "Real World" Barrier)

- **Clean & Impute:** The financial data may contain missing values (NaNs) and formatting inconsistencies (e.g., numbers stored as text with symbols like ₹ or ,). You must clean this and impute missing values intelligently.
- **Merge Disparate Sources:** The datasets come from different organizations. District names do not match perfectly (e.g., *'Gulbarga'* vs *'Kalaburagi'* or *'Jalandhar'* vs *'Jullundur'*). You must build a robust logic to retain as much data as possible.
    - **Hint:** Standard joins will fail. Using string similarity libraries like **fuzzywuzzy**, rapidfuzz, or Levenshtein distance is highly recommended to map these entities automatically.
- Target Calculation: The provided **RBI**(finacial) dataset contains data for multiple quarters and years. You must **sum** the "Bank Credit" and **sum** the "Aggregate Deposits" across all available time periods for each district to obtain the total consolidated figures before calculating the ratio.
  CDR=(Total Sum of Credits/Total Sum of Aggregate Deposits)×100

Phase 2: Predictive Modeling

- Build a Regression Model to predict **CDR**.
- **Constraint:** You **CANNOT** use "Aggregate Deposits" or "Bank Credit" as input features for your prediction. You must use *only* the NFHS social indicators (e.g., Literacy Rate, Clean Fuel Usage, Internet Access).

Phase 3: Additional Insight Generation (Extra Points)

- Identify the **Top 5 Socio-Economic Factors** that have the strongest relation with financial activity(CDR)
- **Anomaly Detection (The "Under-Served" Markets):** Identify 5 districts that your model predicts *should* have high credit activity (based on their good social stats) but actually have very low credit activity

5. Deliverables

1. **The Codebase:** A Jupyter Notebook/Python script showing your cleaning logic, fuzzy merging strategy, and model training.

2. **Prediction File (submission.csv):** This file must contain **ONLY ONE COLUMN**: **Predicted_CDR**

   The number of rows must be **exactly equal** to the number of districts in the **social_test.csv.**

   - Values must be in the **same order** as the test file.
   - No index column.
   - No district names.

- No extra columns.

⚠️ Any deviation from this format will lead to rejection.

6. Evaluation Criteria

Criterion	Weightage	Description
Model Accuracy (Private Test)	**40%**	RMSE and other parameters calculated on a **hidden test set of 100 datapoints** held by the organizers.
Data Wrangling	**30%**	Success rate of merging districts and Data Preprocessing also handling spelling mismatches;
Feature Logic and Code Quality	**15%**	Selection of meaningful social features (vs. dumping all columns).
Additional insights given in phase 3	**15%**	Providing the insights as described in phase 3